Week 2

Arrays and Control Statements

Program 4

Write a Java program to calculate the factorial of a number using a loop.

```
import java.util.Scanner;

class Factorial
{
   public static void main(String args[])
   {
      Scanner sc = new Scanner(System.in);

      int n, fact = 1;

      System.out.print("Enter a number: ");
      n = sc.nextInt();

      for(int i = 1; i <= n; i++)
      {
         fact = fact * i;
      }

      System.out.println("Factorial = " + fact);

      sc.close();
   }
}
```

Program 5

Write a Java program to display the Fibonacci series up to N terms.

```
import java.util.Scanner;

class Fibonacci
{
   public static void main(String args[])
   {
      Scanner sc = new Scanner(System.in);

      int n;
      int a = 0, b = 1, c;

      System.out.print("Enter number of terms: ");
      n = sc.nextInt();

      System.out.println("Fibonacci Series:");

      for(int i = 1; i <= n; i++)
      {
         System.out.print(a + " ");
```

```
            c = a + b;
            a = b;
            b = c;
        }

        sc.close();
    }
}
```

Program 6

Write a Java program to input 10 integers into an array and find the largest, smallest, and average.

```
import java.util.Scanner;

class ArrayOperations
{
    public static void main(String args[])
    {
        Scanner sc = new Scanner(System.in);

        int arr[] = new int[10];
        int largest, smallest;
        int sum = 0;
        double average;

        System.out.println("Enter 10 integers:");

        for(int i = 0; i < 10; i++)
        {
            arr[i] = sc.nextInt();
        }

        largest = arr[0];
        smallest = arr[0];

        for(int i = 0; i < 10; i++)
        {
            if(arr[i] > largest)
            {
                largest = arr[i];
            }

            if(arr[i] < smallest)
            {
                smallest = arr[i];
            }

            sum = sum + arr[i];
        }
```

```
        average = (double)sum / 10;

        System.out.println("Largest Number = " + largest);
        System.out.println("Smallest Number = " + smallest);
        System.out.println("Average = " + average);

        sc.close();
    }
}
```